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Seasonal changes in above- and below-ground non-structural carbohydrates (NSC) in *Spartina alterniflora* in a marsh in Georgia, USA

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ABSTRACT

Spartina alterniflora is the dominant grass appearing in salt marshes along the east coast of the USA. The development of predictive, mechanistic models of *Spartina* has been hindered by the lack of information on below-ground biomass and its dynamics, and in particular the storage of resources that can be used for spring re-growth. We studied the dynamics of non-structural carbohydrates (glucose, fructose, sucrose, and starch) and biomass in 8 different above- and below-ground tissues in *S. alterniflora* over the course of a year in a salt marsh on Sapelo Island, Georgia, USA. We found greater seasonal variability in non-structural carbohydrates in *S. alterniflora* than had been previously reported, with concentrations varying between 3.3% through 17.3% of the total biomass and between 0% and 19.5% of dry weight depending on the type of tissue, with statistical differences between the different tissues. We found that sucrose was the dominant non-structural carbohydrate in above- and below-ground tissue, and that this sugar was likely used for long-term storage during winter months and as a resource for early spring growth. Glucose, fructose, and starch showed less variability, with glucose following changes in above-ground biomass more closely indicating their use as short-term storage. We were unable to develop a coherent carbon budget for the plants largely because of uncertainties in modeled net primary production and heterogeneity in below-ground biomass.

1. Introduction

Salt marshes are highly productive, transitional wetland ecosystems lying at the interface between terrestrial and marine systems. Carbon burial rates in salt marshes are typically high, with global estimates varying up to 1713 g C m^{-2} y^{-1} (mean of 218 ± 24 g C m^{-2} y^{-1}) compared to carbon burial rates in temperate, tropical, and boreal forests between 0.7 and 13.1 g C m^{-2} y^{-1} (Mcleod et al., 2011). Along the eastern coast of the United States, salt marshes are dominated by the halophyte C4 plant, *Spartina alterniflora* (Pennings et al., 2012). The extensive below-ground components of *Spartina* protect the salt marsh from erosion by sea level rise and storms (Knutson, 1988) and also serve as a source of organic matter for microbial biomass (Boschker et al., 1999).

Biomass and height forms of *Spartina* differ along the marsh. The tall height form (> 1 m) grows adjacent to tidal creeks and has a higher productivity than the more inland short form (< 50 cm); the medium height form (50–100 cm) is both intermediate in location and productivity (Mendelssohn, 1979). Although the tall-form *Spartina* inhabits only about 10% of a marsh surface area, it accounts for 30–50% of total marsh production (Gallagher et al., 1980; Giurgevich and Dunn, 1979).

Developing carbon budgets for *Spartina* is complicated by the fact that organic carbon produced through photosynthesis can be respired, used for growth and reproduction, or stored as non-structural carbohydrates (NSC) (Morris et al., 1984). In general, non-structural carbohydrates are among the first products of photosynthesis in plants (Danckwerts and Gordon, 1987; Fulkerson and Donaghy, 2001) and about half of NSC produced from photosynthesis is utilized in respiration (Robson, 1973; Lambers, 1985; Danckwerts and Gordon, 1987). The remaining NSC is either metabolized for growth within a few days (Gordon et al., 1977) or stored for long-term use (Danckwerts and Gordon, 1987; Volenec, 1986; Alberda, 1957; Smith, 1974; Slack et al., 2000) providing energy for regrowth. The main components of NSC in most salt marsh grasses are glucose, sucrose, and fructose (Gorham et al., 1980) and C4 plants are also known to predominantly accumulate starch (Smith, 1973). Different sugars play different roles in plant metabolism. For example, sucrose is known from studies of other plants to be ideal for transport and long-term storage because it is non-reducing and relatively insensitive to metabolism (Arnold, 1968; Lemoine, 2000).

There have been only a handful of studies of NSC and its dynamics in *Spartina*. Lytle and Hull (1980) used radiolabeled carbon to show that

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photosynthate produced late in the growing season was translocated to the rhizomes and used during growth in the following spring, but the form of that photosynthate was not specified. Many studies of NSC in *Spartina* give only total NSC concentrations and examine only total above- and total below-ground biomass, though some studies have measured starch, and others have divided below-ground biomass into roots and rhizomes (Garza et al., 1994; Livingstone and Patriquin, 1981; Gallagher et al., 1984; Colmer et al., 1996).

Without data on the amount and type of carbon being stored for growth in the following season, it is hard to develop a predictive model for above- and below-ground production that can be used over multiple years. Models that do not incorporate below-ground storage will begin to diverge from observations as this material is used to promote growth during the spring months. Our study was aimed at getting a better understanding of how NSC and its different components are distributed within *Spartina* and how this changes over time. Based on literature from other plants, we hypothesize that, for the tall phenotype of *S. alterniflora*, NSC concentrations will be greatest in the below-ground tissues during winter, with sucrose being the main NSC component used for long-term storage, and that this will be reflected in different temporal changes in different NSC components. Our ultimate scientific goal is the creation of a data set to be used in developing a mechanistic model of *Spartina* above- and below-ground production that can be run for multiple years and used to help understand marsh biogeochemical processes.

2. Materials and methods

2.1. Site location

Samples of above- and below-ground biomass were collected from among 6 sites within the domain of the Georgia Coastal Ecosystem Long Term Ecological Research (GCE-LTER) program on Sapelo Island, Georgia, on the southeastern coast of the USA (Supplementary information, Fig. S1). The chosen sites were located within a monotypic area of tall form *Spartina*, close to the Duplin River, and adjacent to the GCE-LTER flux tower (31°26′38.2″ N, 81°17′1.0″ W) which provided meteorological data. Samples were collected monthly between September 2013 and August 2014.

2.2. Sampling and experimental methods

Two cores were collected from locations randomly selected from within 6 long-term sampling sites (Supplementary information, Fig. S1). Above-ground samples were clipped at the ground level and the below-ground samples were collected using a 7.5 cm diameter PVC pipe. Cores were taken to a depth of 30 cm, taking care to preserve above-ground shoots; cores were collected such that there were 1 or 2 above-ground shoots per core. Samples were kept at 4 °C for one night. On the next day, the cores were gently washed with flowing fresh water over an 800 μm sieve to remove mud, then dried in an oven at 60 °C for 3 days. Above-ground tissues were separated into 6 distinct categories; green leaves, green stems, senescing yellow leaves, senescing yellow stems, flowers, and brown leaves and stems representing dead material. Only live roots and rhizomes were used and these were distinguished using the method described by Valiela et al. (1976). Below-ground cores were divided into 2 categories; shallow (0–10 cm) and deep (10–30 cm) parts.

Each component of the plant biomass was weighed, and approximately 4 g of each tissue part was ground to a fine powder using a ball mill (model 8000-D Mixer/Mill, Spex Sample Prep, Metuchen, NJ) until the material passed through a 1 mm screen. Ground samples were stored in airtight scintillation vials.

NSC content in samples was analyzed using a method modified from Zhao et al. (2010). Approximately 16 mg of each dried powder sample were placed in a 1.5 mL micro-centrifuge tube. Sugars in samples were solubilized using 80% ethanol in an 80 °C water bath and the extraction process repeated three times to ensure all solubilized sugars from the samples were collected. The final extract volume was centrifuged for 15 min at 13,500 × *g*. Three replicates of supernatant were incubated with a sequence of enzymes in 96-well plates to quantify glucose, fructose, and sucrose content in each sample. Between extraction of each soluble NSC process, the tube containing supernatant was placed in boiling water to halt any digestion of sugars. Starch, the water-insoluble sugar remaining in the pellet after removal of the supernatant, was solubilized following the procedure specified in Zhao et al. (2010) and then reduced to glucose. The absorbance of each sample was determined photometrically at 340 nm using a 96-Well plate reader (Flexstation3 Benchtop Multi-mode Multiplate Reader, Molecular Devices, Sunnyvale, CA). NSC concentrations were quantified according to standard curves using D-(+)-glucose standards that were run with each plate.

2.3. Statistical analysis

Differences in the amount of NSC as a percentage of dry weight between different tissues in the same core (e.g. differences between green leaves and green stems) were determined by calculating the *Z*-scores using the within-measurement means and standard deviations. Differences were deemed significant at the 99% confidence level ($p < 0.01$). In addition, we calculated the cross correlations using the sample cross correlation function (CCF) between the time series of NSC in the different above- and below-ground plant tissue compartments (using R) to look for potential relationships and lags.

2.4. Net primary production estimation

Net production was estimated using a model that depends on biomass, total irradiance at the top of the canopy, the angle of the sun above the horizon, temperature, ratio of green to total above-ground biomass, and percent nitrogen in dry leaves (Morris et al., 1984):

$$P = \frac{\psi TNF \sin(\vartheta)\{\ln(Le^{\alpha B_c/\sin(\vartheta)} + \lambda) - \ln(L + \lambda)\}}{(\alpha(N + \eta))} - \rho T(FB_c + B_b) \quad (1)$$

where the symbols and values for constants are given in Table 1. Total irradiance was estimated from PAR measured at the flux tower (Supplementary information, Fig. S2) assuming that PAR was 50% of total irradiance and that there were 2.77×10^{18} quanta s^{-1} W^{-1} (Kirk, 1994); measured PAR was available as 5-minute data-logged values.

Table 1
Definitions of symbols and parameter values. Parameter values were taken from Morris et al. (1984).

Symbol	Name	Units and value
P	Total net production	g dw m^{-2} h^{-1}
B_c	Above ground biomass	g dw m^{-2}
B_b	Below ground biomass	g dw m^{-2}
T	Air temperature	°C
ϑ	Solar elevation	Radians
L	Solar irradiance at the top of canopy	W m^{-2}
λ	Half saturation constant for irradiance	300 ± 100 W m^{-2}
α	Irradiance extinction rate within canopy	(3.4 ± 1.0) × 10^{-4} m^2 g dw^{-1}
N	Percent nitrogen in dry leaves	%
F	Ratio of green tissue to total canopy biomass	Dimensionless
ψ	Temperature coefficient for gross production	(7.1 ± 1.7) × 10^{-4} $°C^{-1}$ h^{-1}
η	Half saturation constant for nitrogen	0.36 ± 0.29% dw^{-1}
ρ	Temperature coefficient for dark respiration	(2.3 ± 0.6) × 10^{-5} $°C^{-1}$ h^{-1}

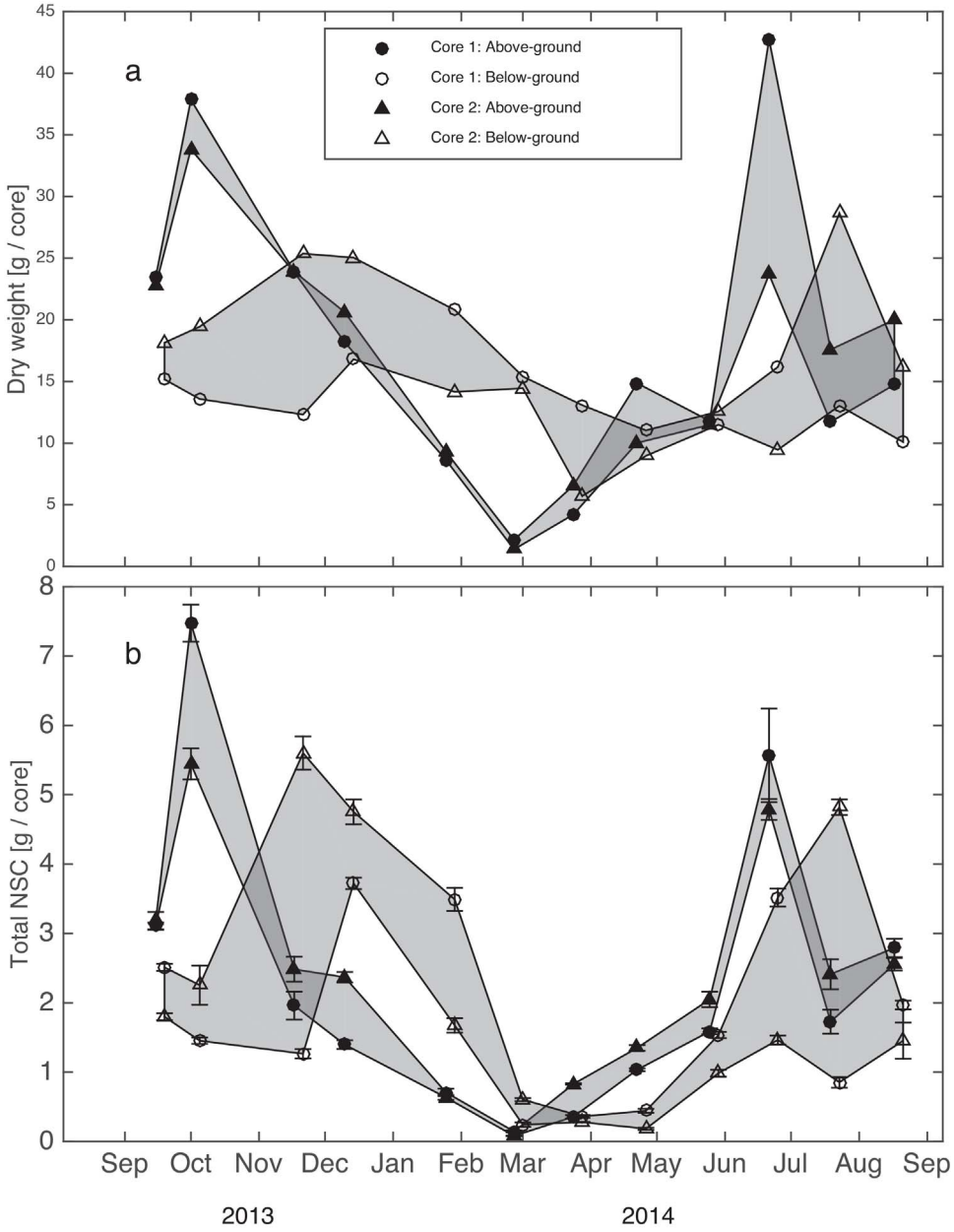


Fig. 1. Measured dry weight and non-structural carbohydrates in each core. (a) Above- and below-ground dry mass. (b) Amount of non-structural carbohydrate from triplicate measurements of the same core (n = 3), error bars represent standard deviation. Shaded areas are guides to indicate when measurements between the cores overlap. Values for core 1 and core 2 are offset by 2 days from the actual day of measurement for clarity (both cores were taken on the same day).

The angle of the sun above the horizon was calculated using standard equations (Kirk, 1994) and air temperature was measured at the GCE Flux Tower, a level 3 weather station, adjacent to the sampling site. Above- and below-ground areal biomasses were calculated on a per meter square basis by making use of stem density measurements made in adjacent plots along a transect near the flux tower at the same time as the cores were taken; above- and below-ground biomasses measured in the sample cores were scaled up assuming an average of 2 stems per core. Daily values of biomass were obtained by fitting a piecewise cubic Hermite interpolating polynomial to the monthly data. The ratio of green to total above-ground biomass was calculated from the core data as well—daily values were calculated using the same technique as for biomass. Nitrogen concentrations in above-ground components of *Spartina*, including the brown, yellow and green tissues, were measured next to our plots by colleagues (unpubl. results). This nitrogen data showed a slightly lower concentration (~0.3%) than the data from Morris et al. (1984) which measured nitrogen in only green leaves; both data sets show the same seasonal pattern. We have compared model results using the new nitrogen data with those using Morris' data and found no significant change (mean = 0.03, variance = 0.007) in the difference of the two results in the model. Consequently, the percent nitrogen in leaves was assumed to be same as that given in Morris et al. (1984). All computer code was written and run using Matlab 2016b.

3. Results

3.1. Environmental conditions

Solar irradiance and air temperature showed the expected seasonal cycles (Supplementary information, Fig. S2). The minimum daily average temperature was generally above 0 °C except for a brief period during January 2014. Daily precipitation was less than about

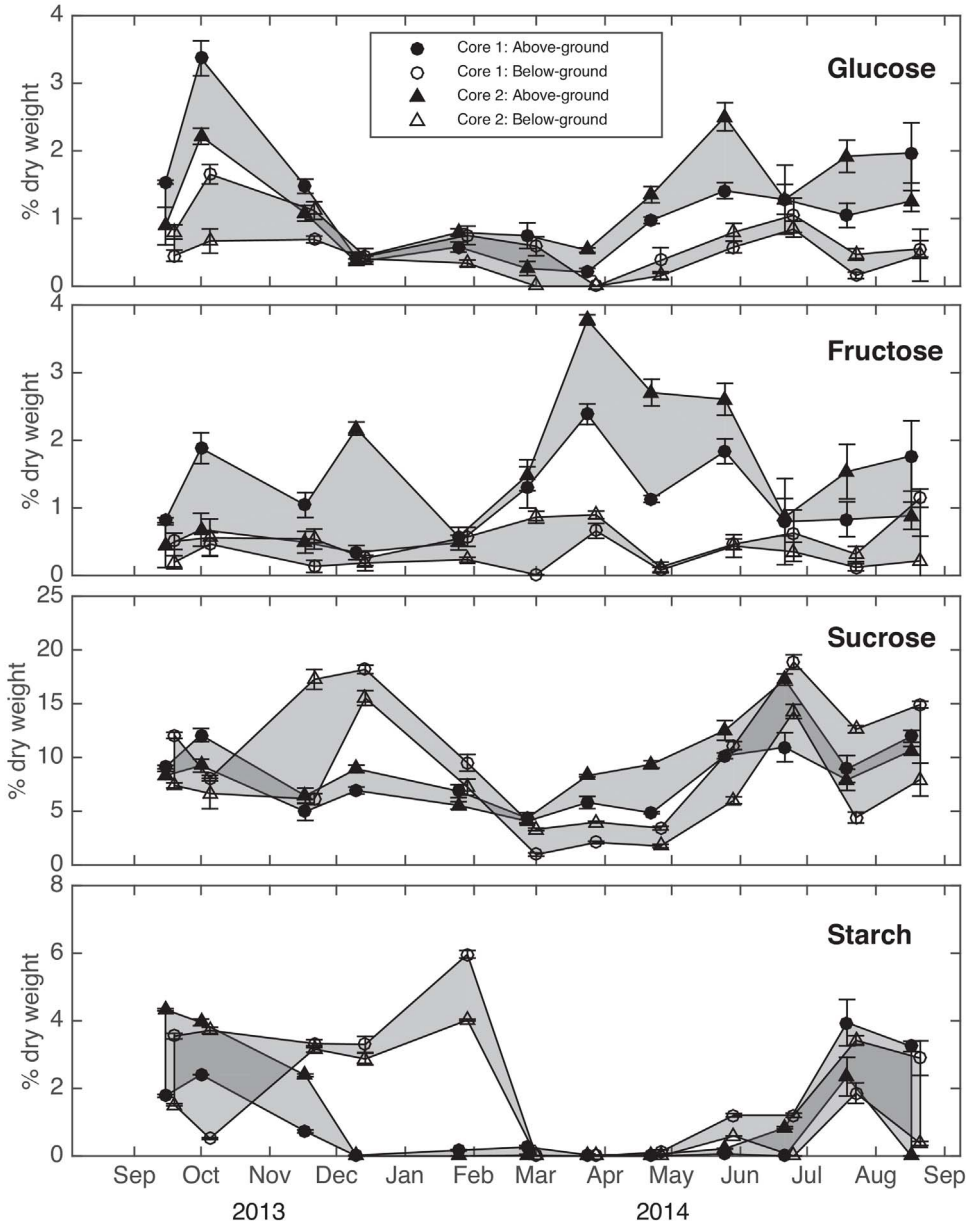


Fig. 2. Above- and below-ground glucose, fructose, sucrose, and starch concentrations as percentages of above- and below-ground dry weight combined from triplicate measurements of the same core (n = 3); error bars represent standard deviation. Shaded areas are guides to indicate when measurements between the cores overlap. Values for core 1 and core 2 are offset by 2 days from the actual day of measurement for clarity (both cores were taken on the same day).

10.5 mm d^{-1} and over the year was typical for the location (Supplementary information, Fig. S2).

3.2. Biomass

The total above-ground biomass showed a typical, strong annual cycle with maximum biomass occurring in late summer through early fall and a minimum in February (Fig. 1a). Spring growth started to occur in early spring, with the initial greening of above-ground biomass occurring in March. On the whole, the total below-ground biomass was approximately constant within measurement uncertainty with the highest values occurring in December (20.9 ± 6.2 g $core^{-1}$) and July (20.8 ± 9.0 g $core^{-1}$) and the lowest in March (9.4 ± 3.7 g $core^{-1}$). The total above-ground biomass leads the below-ground biomass with a 1-month lag (ρ = 0.91).

3.3. Non-structural carbohydrates

Total non-structural carbohydrates show distinct above- and below-ground patterns. Unsurprisingly, above-ground NSC follows a similar seasonal pattern to above-ground biomass (Fig. 1). Below-ground total NSC shows distinct patterns suggestive of carbon net primary production and translocation (Fig. 1b). Below-ground total NSC peaks in mid-winter (4.4 ± 1.5 g $core^{-1}$ in December) as above-ground biomass and NSC are declining—there is a suggestion of an increase in below-ground biomass, but this is well within the measurement uncertainties (Fig. 1a). Below-ground total NSC drops approximately an order of magnitude to a minimum of 0.44 ± 0.12 g $core^{-1}$ in late February, remaining approximately constant at these low levels until April and then increasing to 2.9 ± 1.7 g $core^{-1}$ in July.

The dynamics of the four measured NSC differ between above- and below-ground tissues. Glucose and fructose were generally larger, as a percentage of dry mass, in above-ground tissues (Fig. 2). Sucrose

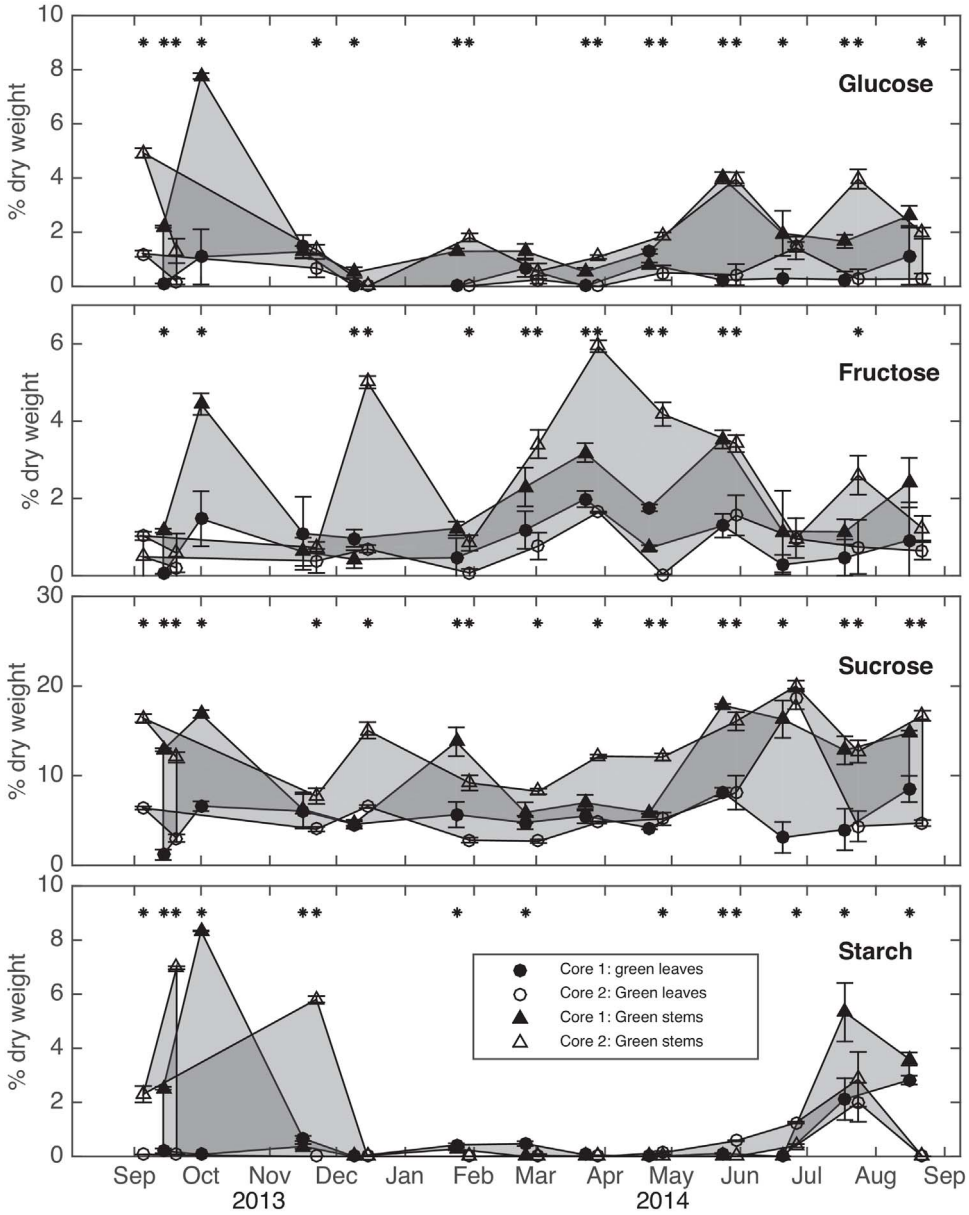


Fig. 3. Within-core measurements of glucose, fructose, sucrose and starch as a percentage of dry weight in green stems, green leaves (n = 3). Error bars represent standard deviation. Asterisks indicate significant differences (p = 0.01, n = 3) between green leaves and green stems for each individual core. Shaded areas are guides to indicate when measurements between the cores overlap. Values for core 1 and core 2 are offset by 2 days from the actual day of measurement for clarity (both cores were taken on the same day).

concentrations were generally similar between above- and below-ground tissues, whereas starch concentrations were greater in below-ground tissues during the winter (Fig. 2). Glucose concentrations showed broadly similar temporal patterns to above-ground biomass, with peaks in above-ground biomass occurring in October and late May (Figs. 1 and 2). Sucrose was generally the dominant form of NSC in both above- and below-ground tissues, with concentrations in above-ground tissue varying between approximately 5% and 14% of dry weight and between approximately 1% and 17% of dry weight in below-ground tissue. Starch concentrations showed distinctive temporal patterns (Fig. 2). Above-ground starch concentrations were high in late autumn, dropping to almost zero in December before increasing again in early summer. Below-ground concentrations were between approximately 2% and 5% of dry weight until March when they dropped to zero before increasing again in June.

The dynamics of the different NSC components also varied between different tissue types. Changes in glucose and fructose showed similar patterns in green leaves and stems (Supplementary information, Table S3, Fig. 3), with 2 peaks in October and early summer. Both sugars were a generally higher proportion of dry weight, and showed greater temporal variability in green stems. Similarly, sucrose concentrations in green leaves and green stems showed very similar temporal patterns, with concentrations in green stems being generally higher than in green leaves. Interestingly, the peak in sucrose concentrations occurs a month later than the peak in glucose and fructose concentrations. Starch was almost absent from green leaves except during the summer. However, starch in green stems showed a distinctive pattern with peaks in October and July and being zero or less than 1% between December and May. Although there is a lot of scatter in the data and differences between cores, NSC in green leaves and stems were for the most part statistically different within individual cores throughout the year.

Glucose and fructose in yellow leaves and stems were generally lower than in green leaves and stems (Fig. 4). Glucose concentrations were generally higher in the fall and winter, with values typically < 0.5% dry weight throughout the spring—a similar pattern was seen in sucrose concentrations, though concentrations in the fall and winter

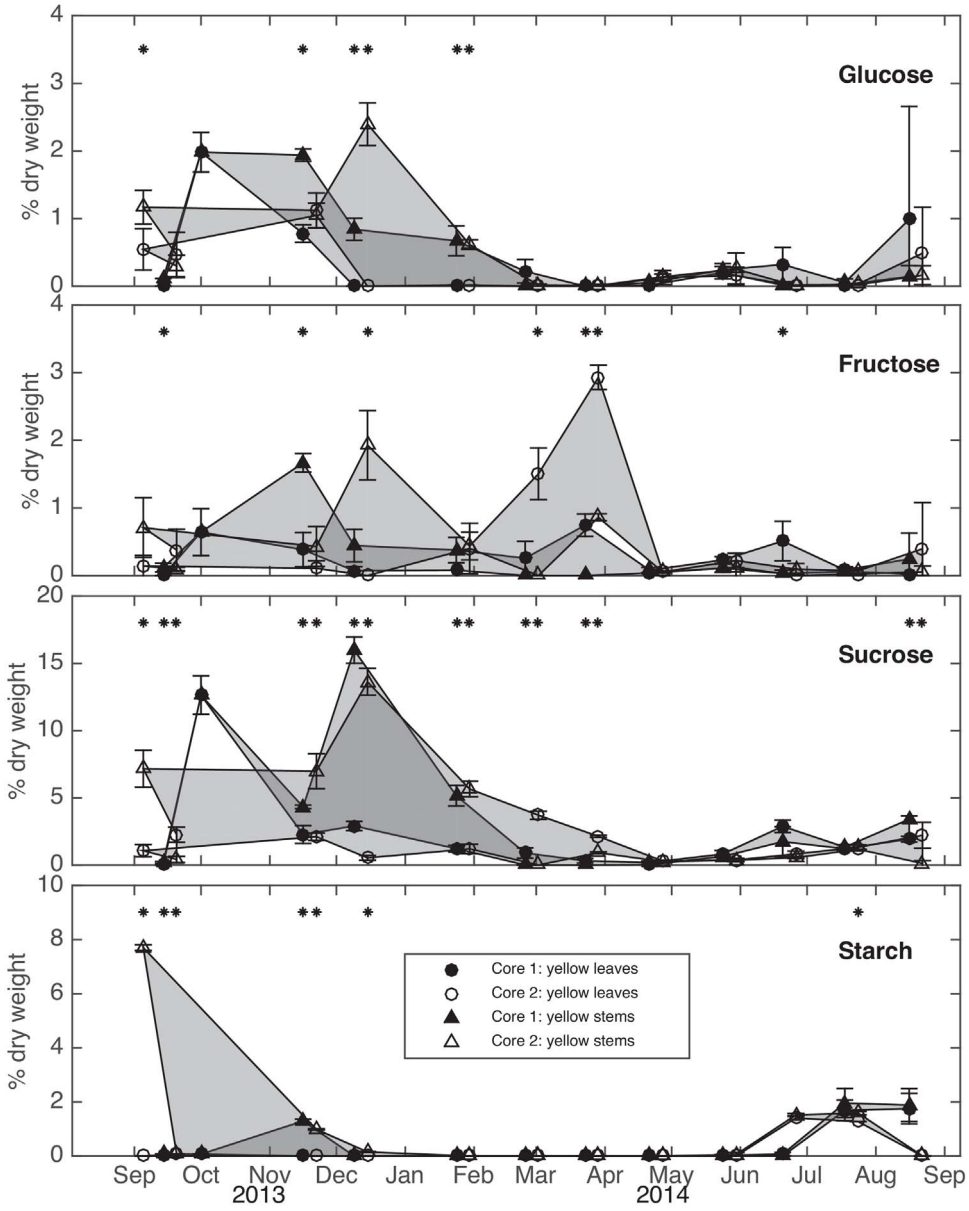


Fig. 4. Within-core measurements of glucose, fructose, sucrose and starch as a percentage of dry weight in yellow stems and yellow leaves (n = 3). Error bars represent standard deviation. Asterisks indicate significant differences (p = 0.01, n = 3) between green leaves and green stems for each individual core. Shaded areas are guides to indicate when measurements between the cores overlap. Values for core 1 and core 2 are offset by 2 days from the actual day of measurement for clarity (both cores were taken on the same day).

were higher than those for glucose. Fructose concentrations tended to remain slightly higher throughout the early spring, decreasing in April to < 1% dry weight. Within-core concentrations of both fructose and sucrose were generally statistically different between yellow stems and yellow leaves; some differences were seen for glucose when concentrations were relatively high in the fall and winter, but within-core yellow leaf and yellow stem concentrations were statistically similar for the lower concentrations found in the spring and summer. The seasonal pattern for starch in yellow leaves and stems was similar to that for green leaves and stems (Figs. 3 and 4). Non-structural carbohydrates were extremely low or below the detection level in brown leaves and stems and flowers (not shown).

Concentrations of glucose, sucrose, and starch in below-ground tissue tended to show minimum values in late spring or early summer, whereas fructose concentrations were approximately constant (Fig. 5). Sucrose was the dominant with maximum concentrations of approximately 20% dry weight, similar to those seen in green leaves and stems. In general, within core NSC concentrations in 0–10 cm below-ground tissue were statistically different from those in the 10–30 cm depth range. However, there was considerable below-ground heterogeneity as depicted by the increase in non-overlapping patches in Fig. 5. What is more, within-core measurements sometimes showed greater NSC concentrations in the 0–10 cm that the 10–30 cm depth range, while this was reversed at other times; such reversals did not appear in green or yellow tissues (Figs. 3 and 4).

Maximum concentrations of NSC were found in above- and below-ground tissues at different times of the year. Green stems had the highest concentration (measured as a percent of dry weight) of NSC at the start of the sampling in September 2013, and again between March through June of 2014. Maximum NSC concentrations showed a progression from green stems, through yellow stems, to below-ground biomass (Table 2) suggestive of a time progression of storage.

3.4. Net primary production

Estimates of net primary production show an expected seasonal

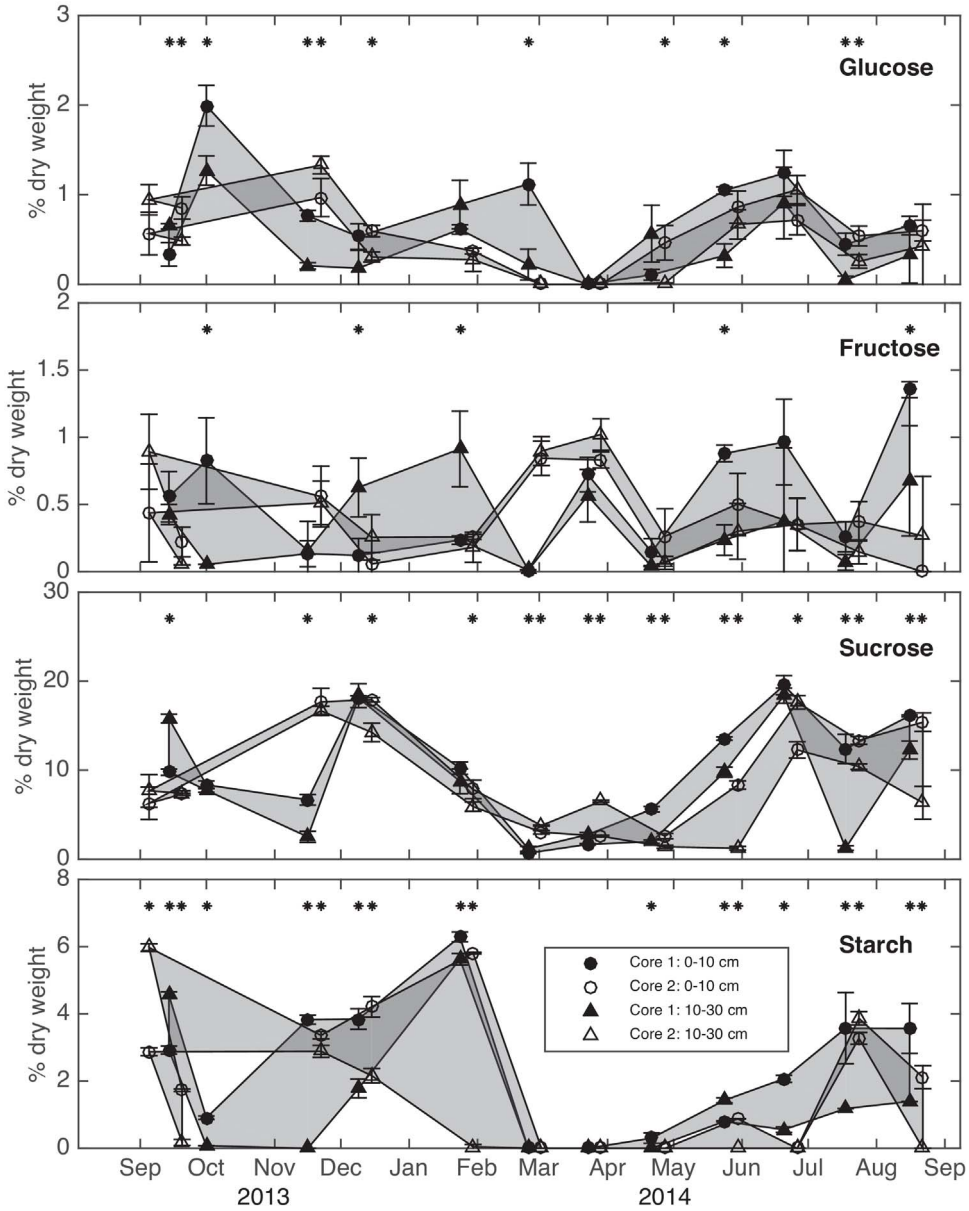


Fig. 5. Within-core measurements of glucose, fructose, sucrose and starch concentrations as percentages of dry weight in below-ground tissue, divided in to shallow (0–10 cm) and deep (10–30 cm) zones ($n = 3$). Error bars represent standard deviation. Asterisks indicate significant differences ($p = 0.01$, $n = 3$) between green leaves and green stems for each individual core. Shaded areas are guides to indicate when measurements between the cores overlap. Values for core 1 and core 2 are offset by 2 days from the actual day of measurement for clarity (both cores were taken on the same day).

Table 2
Distribution of the total NSC as a percentage of dry weight within different tissues of *Spartina alterniflora*. Values obtained by combining measurements from the two cores.

	Green leaves (%)	Green stems (%)	Yellow leaves (%)	Yellow stems (%)	Belowground 0–10 cm (%)	Belowground 10–30 cm (%)	Brown leaves and stems (%)	Flowers (%)	Total biomass (%)
Sep. 2013	2.70	57.16	0.03	0.24	28.37	11.51	0.00	0.00	13.04
Oct. 2013	8.67	37.94	1.03	29.08	14.37	8.64	0.07	0.20	15.82
Nov. 2013	2.64	20.53	0.65	19.52	42.09	14.33	0.17	0.06	11.98
Dec. 2013	3.31	13.05	0.74	12.57	38.49	31.85	0.00	0.00	15.33
Jan. 2014	3.16	12.68	0.61	5.35	50.93	26.65	0.63	n.p	10.09
Feb. 2014	9.87	7.91	2.40	n.p	39.36	40.45	n.p	n.p	3.28
Mar. 2014	20.54	39.05	0.67	0.10	18.34	21.30	n.p	n.p	6.20
Apr. 2014	20.28	59.79	0.06	0.10	11.23	8.55	n.p	n.p	6.74
May. 2014	17.95	41.18	0.13	0.17	27.43	13.14	n.p	n.p	13.00
June. 2014	17.80	50.42	0.57	0.45	14.96	15.80	n.p	n.p	17.28
July. 2014	9.24	31.98	0.28	0.33	43.82	14.35	n.p	n.p	13.85
Aug. 2014	9.65	47.80	0.95	0.20	21.54	19.64	0.23	n.p	14.84

n.p.: not present.

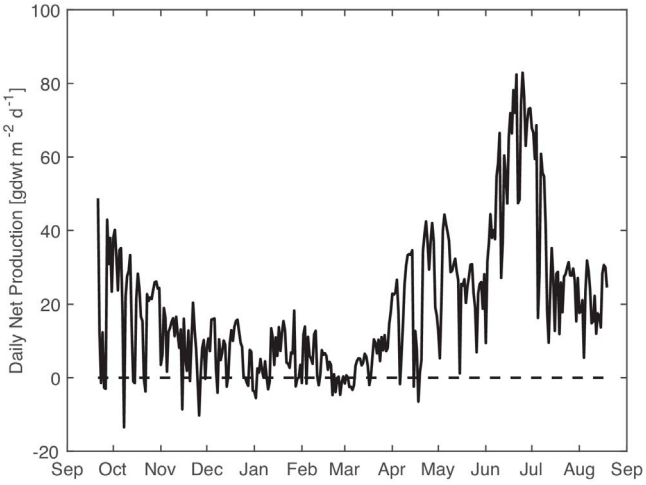


Fig. 6. Estimated daily net production using the Morris model (Eq. (1)) and local environmental variables (Supplementary information, Fig. S2).

cycle, with values in late February and early March being consistently negative (Fig. 6), corresponding the times of lowest above-ground biomass (Fig. 1) and lowest sucrose and starch concentrations in above- and below-ground tissue (Fig. 2). Estimates of net primary production reached maximum values of 88 g dw(dry weight) $m^{-2}\,d^{-1}$ during the summer months. Annual estimates of below-ground gross and net production and below-ground respiration were made using the same calculations as found in Morris et al. (1984). Below-ground gross production was estimated to be 4.2×10^3 g C $m^{-2}\,y^{-1}$ and below-ground respiration was 1.3×10^3 g C $m^{-2}\,y^{-1}$.

4. Discussion

4.1. Differences in NSC Dynamics

The dynamics of above- and below-ground NSC components were quite different, indicating that different components are used for different purposes. Total NSC in below-ground tissue showed an expected seasonal pattern consistent with a significant component of NSC being translocated to the below-ground tissue where it was stored and used for metabolism during the winter and to help with regrowth in the spring. Most of this change appeared to be a result of changes in below-ground sucrose concentrations, which made up ~50–90% of the below-ground NSC. Starch also showed significant changes in below-ground tissue, but these were far more variable and harder to interpret (Fig. 5). Below-ground glucose concentrations also declined during fall and winter, but results were very variable and accounted for, at most, 2% of dry weight. Previous results by Lytle and Hull (1980) suggest that photosynthate is translocated below-ground and used for growth in the following spring. Our results are consistent with this finding and suggest that sucrose is mainly used for this purpose. Non-structural carbohydrate content in above-ground tissue peaked in fall and summer, with a distinct minimum in early spring. Glucose tends to be used in plants for short term energy storage and use, so we would expect it to show seasonal changes.

Our results are in general agreement with previous measurements but shed additional light on the distribution of the different sugars within the plant over time. Live above- and below-ground biomass showed similar annual variation to that found for *Spartina* at Sapelo by Schubauer and Hopkinson (1984). Previous measurements of NSC in *Spartina* from Georgia salt marshes showed NSC accounting for 4–10% of biomass (McIntire and Dunstan, 1976), whereas values between 9 and 31% were found for *Spartina* growing in Nova Scotia, Canada (Livingstone and Patriquin, 1981). In our samples, total NSC concentration varied between 3 and 17% of dry weight, a slightly larger range than that seen by McIntire and Dunstan (1976) but still lower than values seen in Nova Scotia. Studies transplanting short form *Spartina* from different regions suggest that such a variation may be genetically determined and a result of *Spartina* from higher latitudes storing more NSC in order to survive under harsher winter conditions (Seliskar et al., 2002). In addition, the distribution of the total NSC as a percentage of dry weights within different tissues of *Spartina* showed an interesting pattern where by the total NSC is stored most in above-ground parts (Table 2) during the fall but its main distribution moves toward the below-ground in winter and returns back to the above-ground as the spring seasons approach.

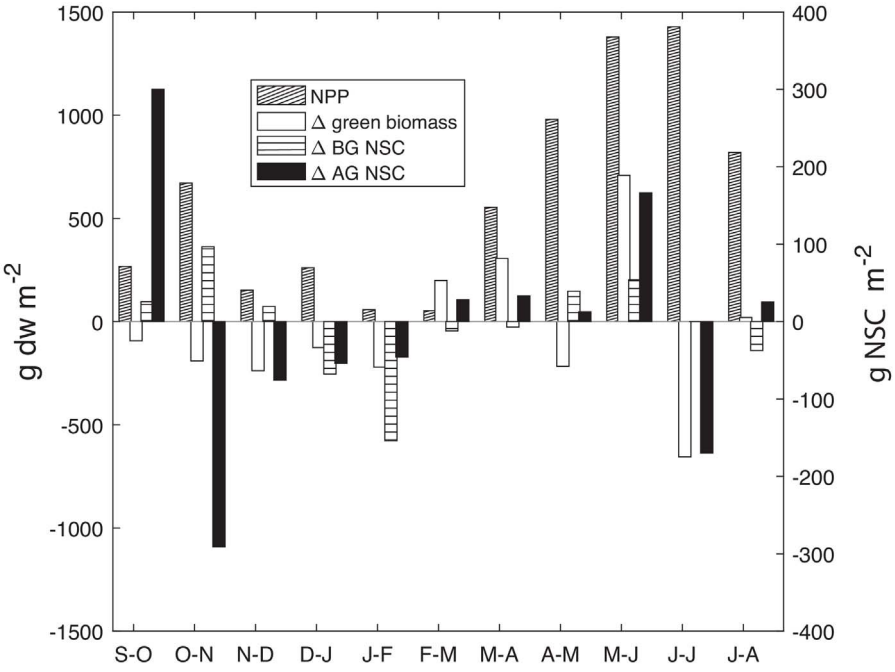


Fig. 7. Comparison of monthly net primary production (estimated from the model, Eq. (1)) with observed monthly changes in green biomass, below-ground non-structural carbohydrate and above-ground non-structural carbohydrate.

4.2. Consequences for modeling of Spartina

Our main scientific objective for this study was to develop a data set that would inform the development of a model of above- and below-ground production of *Spartina*. Predictive models of *Spartina* production rely on accurately determining the net balance between the production of new biomass and the loss of old biomass. Current models for *Spartina* do not account for the storage and subsequent use of non-structural carbohydrates. Such models may under-predict year-to-year biomass if, as has been suggested, this material is used to maintain biomass during conditions of low primary production and to provide energy for initial production of photosynthetic biomass in the spring.

Morris et al. (1984) used a model to predict annual total NPP for tall *Spartina* on Sapelo Island of 2200 g $C\,m^{-2}\,y^{-1}$, split between 800 g $C\,m^{-2}\,y^{-1}$ above-ground and 1400 g $C\,m^{-2}\,y^{-1}$ below-ground. Total NPP over our sampling period, September 2013 to August 2014, estimated using the model given in Morris et al. (1984) was 2608 g $C\,m^{-2}\,y^{-1}$. A carbon concentration of 42.9% g dw (Morris et al., 1984) was used to convert from dry weight biomass to carbon since our observed carbon concentration data (unpublished) showed a similar value (41 ± 0.6%) throughout the seasons. However, the Morris *et al.* model does not allow for loss of above-ground tissue through grazing, the assumption being that loss of above-ground tissue occurs when respiration exceeds gross production. However, Fig. 7 shows months (e.g. October–November) when there was a decrease in above-ground green tissue and a simultaneous positive total NPP. This period corresponded to a period of decrease in total above-ground biomass (Fig. 1). This may also be a result of the equation for NPP not including yellow tissue in calculating the respiration. The high NPP from July to August (Fig. 7) may be partially explained by the equation for NPP not including increased transpiration, energy consumed for osmotic adjustment (Hessini et al., 2009) or increased dark respiration (Giurgevich and Dunn, 1979).

We were unable to develop a consistent carbon budget for the whole plant. For some months the plants appear to be in a state of approximately balanced growth (e.g. September to October), but in others, there is an excess of estimated net primary production (e.g. December to January). There are several possible causes for this. One is that there are processes missing from the model we used to estimate NPP. For example, loss of above-ground biomass due to herbivory is not included in the model. Another possibility is that sampling was not representative of the actual biomass. As discussed below, this is always a problem in dealing with strongly heterogeneous environments such as salt marshes.

S. alterniflora is a clonal plant with significant below-ground biomass. Above-ground annual primary production ranges between 150 g $m^{-2}\,y^{-1}$ and 900 g $m^{-2}\,y^{-1}$ (Kirwan et al., 2009) and below-ground net production between 600 and 1400 g $C\,m^{-2}\,y^{-1}$ with an estimated translocation between 1200 and 6800 g $C\,m^{-2}\,y^{-1}$ of above-ground production to below-ground tissues (Morris et al., 1984). The heterogeneity of the below-ground biomass makes a precise and accurate sampling of this material something of a problem. NSC measurements from replicate cores differed sometimes by a factor of 2. We can estimate the number of cores that would have had to be taken to achieve an accuracy of 1 mg g dw^{-1} in determining NSC. If we assume that the parent distribution is a normal distribution with a range of approximately 4 times the population standard deviation (Thomson and Emery, 2014), then we estimate that typically we would need of the order of 20 cores per time point to achieve this kind of precision. This not only constitutes a considerable investment of labor but would also significantly alter the salt marsh and suggests that non-destructive methods (e.g. O'Connell et al., 2015) may provide better approaches for at least assessing below-ground biomass.

There was significant between-core variability in NSC concentrations, especially for below-ground material. Differences between the 0–10 cm and 10–30 cm depth ranges were generally significant ($p < 0.01$). However, the maximum concentration of NSC could occur in either depth range. Patterns of above-ground NSC concentrations were more clearly delineated; for example, green stems generally had greater NSC concentrations than green leaves. We suggest that this heterogeneity in below-ground properties, along with the relatively constant below-ground biomass, argues for considering below-ground biomass as a single component in models, rather than subdividing it.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.aquabot.2017.04.003.

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